IGCSE 0607 Practice Questions — E1.5 — Combined Questions & Markscheme

Ordering

Total Questions: 18 — **Total Marks:** 78 Marks

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1 Insert one of the symbols $<$, $>$ or $=$ to make each statement correct.

(i) $0.45 \ldots\ldots \dfrac{2}{5}$

(ii) $-3 \ldots\cdots -7$

[2]

2 Write the following in order of size, starting with the smallest.

$$\frac{5}{8} \qquad 0.62 \qquad 61.5\% \qquad \frac{3}{5}$$

..... **[2]**

3 Write $<, >$ or $=$ to make each statement true.

(a) $0.45 \ldots\ldots\ldots \frac{4}{9}$

[1]

(b) $2^{-3} \ldots\ldots\ldots 0.125$

[1]

(c) $-1.5 \ldots\ldots\ldots - \frac{7}{5}$

[1]

4 **(a)** Insert one of the symbols $<, =$ or $>$ to make each statement true.

(i) $\frac{5}{8} \ldots 0.65$

[1]

(ii) $72\% \ldots \frac{18}{25}$

[1]

(b) Write the following in order of size, starting with the smallest.

$$0.55 \qquad \frac{6}{11} \qquad 54\% \qquad \frac{5}{9}$$

..... **[2]**

5 **(a)** Write the following numbers in order of size, starting with the smallest.

$$\frac{3}{4} \qquad 0.72 \qquad 74\% \qquad \frac{5}{7} \qquad 0.7\dot{4}$$

..... **[2]**

(b) Complete each statement by inserting the correct symbol from $=$, $>$, or $<$.

(i) $-\frac{5}{8}$ ..... -0.65

[1]

(ii) $(0.3)^2$ 9%

[1]

6 **(a)** Insert $<$, $>$ or $=$ to make each statement correct.

(i)

$$0.4 \ldots\ldots \frac{2}{5}$$

[1]

(ii)

$$-7 \ldots\ldots -9$$

[1]

(iii)

$$2^3 \ldots\ldots \sqrt{25}$$

[1]

(b) Write these in order of size, starting with the smallest.

$$33\% \qquad \frac{1}{3} \qquad 0.3 \qquad \frac{3}{8}$$

..... **[2]**

7 **(a)** Given that $0 < p < 1$, write the following expressions in order of size, starting with the smallest.

$$p \qquad p^2 \qquad \sqrt{p} \qquad \frac{1}{p}$$

..... **[2]**

(b) Given that $-1 < q < 0$, write $<, >$ or $=$ to make each statement true.

(i) $q^3 \ldots\ldots\ldots q$

[1]

(ii) $\frac{1}{q} \ldots\ldots\ldots q$

[1]

8 **(a)** Given that $0 < x < 1$, write the following expressions in order of size, starting with the smallest.

$$x \qquad x^2 \qquad \frac{1}{x} \qquad \sqrt{x}$$

..... **[2]**

(b) Given that $x < -1$, write the following expressions in order of size, starting with the largest.

$$x \qquad x^3 \qquad x^{-2} \qquad \frac{1}{x}$$

..... **[2]**

9 **(a)** Given that $x = -0.4$, write the following in order of size, starting with the smallest.

$$x \qquad x^2 \qquad x^3 \qquad \frac{1}{x}$$

..... **[2]**

(b) Insert the correct symbol ($<, =$, or $>$) to make the statement true.
Explain your reasoning.

$5\sqrt{3} \ldots 4\sqrt{5}$

[2]

(c) Find the integer m that satisfies the inequality

$$\frac{m}{7} < \frac{5}{9} < \frac{m+1}{7}$$

$m =$ ..... **[3]**

10 **(a)** It is given that $0 < p < 1$.
Write the following values in order of size, starting with the smallest.

$$p \qquad \sqrt{p} \qquad p^2 \qquad \frac{1}{p}$$

..... **[2]**

(b) Write the correct symbol $<$, $>$ or $=$ to complete each logical statement.

(i) If $x < 0$, then $x^3 \ldots\ldots x^2$.

[1]

(ii) If $x > 1$, then $x^{-1} \ldots\ldots 1$.

[1]

(c) Given $a = 2^{50}$, $b = 3^{30}$ and $c = 5^{20}$, write a, b and c in order of size, starting with the smallest.
You must show all your working.

..... **[3]**

11 Let x be a real number such that $0 < x < 1$.

(a) Write the following four expressions in order of size, starting with the smallest.

$$x^2 \qquad \sqrt{x} \qquad \frac{1}{x} \qquad x$$

..... **[2]**

(b) Now let k be a negative integer such that $k \leq -2$.
Write the following four expressions in order of size, starting with the smallest.

$$k^3 \qquad \frac{1}{k} \qquad k^2 \qquad -\frac{1}{k^2}$$

..... **[3]**

(c) Find the range of values of n for which $n^2 \leq n$ and $n \neq 0$.
Express your answer as an inequality.

..... **[3]**

12 Write the following in order of size, starting with the smallest.

$$\frac{4}{33} \qquad 0.12 \qquad 12.1\% \qquad \sqrt{0.015}$$

..... **[2]**

13 Write the following masses in order of size, starting with the largest.

5.2×10^{-4} kg 0.04 g 4.5×10^{3} mg 6×10^{-7} tonnes

..... **[3]**

14 **(a)** Here are four numerical expressions.

$a = 3.1 \times 10^3$
$b = \sqrt{9.8 \times 10^6}$
$c = 5^5$
$d = 56^2$

Work out the value of each expression and write them in order of size, starting with the largest.

..... **[3]**

(b) State the integer n such that

$$n < \sqrt{250} < n + 1$$

$n =$ ..... **[1]**

15 **(a)** Write these numbers in order of size, starting with the smallest.

$$5^{-3} \qquad 8.5 \times 10^{-3} \qquad 0.00805 \qquad \frac{1}{120}$$

..... **[2]**

(b) The mass, m kilograms, of a bag of rice is 12 kg, correct to the nearest whole kilogram.
Write down the inequality representing the possible values of m.

..... **[2]**

16 **(a)** Write the following numbers in order of size, starting with the smallest.

$$\frac{7}{8} \qquad 86\% \qquad 0.865 \qquad \frac{5}{6}$$

..... **[2]**

(b) Write down a fraction, f, such that $\frac{1}{3} < f < \frac{1}{2}$.

..... **[2]**

17 Consider the five numbers p, q, r, s and t defined below:

$p = 4.8 \times 10^{-3}$

$q = \dfrac{1}{205}$

$r = \sqrt{2.3 \times 10^{-5}}$

$s = 0.485\%$

$t = (0.07)^2$

(a) Write the five numbers p, q, r, s and t in order of magnitude, starting with the largest.

..... **[3]**

(b) The number x satisfies the inequality $r < x \leq p$.
Write down a possible value for x in standard form.

$x =$ ..... **[1]**

(c) Given that $y \neq t$ and $y \geq q$, determine whether the statement $y > s$ is always true, sometimes true, or never true.
Explain your answer.

.....

..... **[2]**

18 **(a)** Given that $x > 0$ and $y < 0$, insert the correct symbol $<$, $>$ or $=$ to make the statement correct.

$x^2y \ldots\ldots 0$

[1]

(b) Write these expressions in order of size, starting with the smallest, when $a = 0.5$.

$$a^2 \qquad a \qquad \sqrt{a} \qquad \frac{1}{a}$$

..... **[2]**

(c) State whether the following statement is True or False.
Give a reason for your answer.
If $p > q$, then $\frac{1}{p} < \frac{1}{q}$ for all non-zero real numbers p and q.

..... **[2]**

1 (i) $>$ **[B1]**

(ii) $>$ **[B1]**

[Back to Question] *[Total Marks: 2]*

2 Converts to a common format (e.g. decimals: $0.625, 0.62, 0.615, 0.6$) **[M1]**

$\frac{3}{5}, 61.5\%, 0.62, \frac{5}{8}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

3 **(a)** $>$ **[B1]**

(b) $=$ **[B1]**

(c) $<$ **[B1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** **(i)** $<$ **[B1]**

(ii) $=$ **[B1]**

(b) M1 for converting at least three values into a comparable form (e.g. percentages: 55%, $54.54\%\ldots$, 54%, $55.55\%\ldots$) **[M1]**

$54\%, \frac{6}{11}, 0.55, \frac{5}{9}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** Converting to comparable decimals: 0.75, 0.72, 0.74, $0.714\ldots$, $0.744\ldots$ (at least three correct) **[M1]**

$\frac{5}{7}$, 0.72, 74%, $0.7\dot{4}$, $\frac{3}{4}$ **[A1]**

(b) **(i)** $>$ **[B1]**

(ii) $=$ **[B1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** **(i)** $=$ **[B1]**

(ii) $>$ **[B1]**

(iii) $>$ **[B1]**

(b) $0.3 \quad 33\% \quad \frac{1}{3} \quad \frac{3}{8}$ **[B2]**

If 0 scored, allow [M1] for converting at least two of the values correctly to decimals, e.g. 0.33, $0.333\ldots$, 0.375

[Back to Question] *[Total Marks: 5]*

7 **(a)** $p^2 \quad p \quad \sqrt{p} \quad \frac{1}{p}$ **[B2]**

M1 for evaluating the expressions with a chosen test value for p (e.g., $p = 0.25$) OR for any 3 expressions in the correct relative order.

(b) **(i)** $>$ **[B1]**

(ii) $<$ **[B1]**

[Back to Question] *[Total Marks: 4]*

8 **(a)** $x^2, x, \sqrt{x}, \frac{1}{x}$ **[B2]**

(B1 for 3 in correct relative order or if reversed)

(b) $x^{-2}, \frac{1}{x}, x, x^3$ **[B2]**

(B1 for 3 in correct relative order or if reversed)

[Back to Question] *[Total Marks: 4]*

9 **(a)** M1 for correctly calculating at least two values: $x^2 = 0.16$, $x^3 = -0.064$, $\frac{1}{x} = -2.5$ **[M1]**

$\frac{1}{x}, x, x^3, x^2$ **[A1]**

(b) M1 for squaring both terms to get 75 and 80, OR expressing as $\sqrt{75}$ and $\sqrt{80}$ **[M1]**

$<$ **[A1]**

(c) M1 for identifying a common denominator, e.g. cross multiplying to get $9m < 35$ **[M1]**

M1 for $35 < 9(m + 1)$ leading to $26 < 9m$ **[M1]**

3 **[A1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** $p^2 \quad p \quad \sqrt{p} \quad \frac{1}{p}$ **[B2]**

If 0 scored, allow [M1] for a correct order of any three of the terms or for successfully substituting a test value such as $p = \frac{1}{4}$ to evaluate all four expressions.

(b) **(i)** $<$ **[B1]**

(ii) $<$ **[B1]**

(c) c, b, a or $5^{20}, 3^{30}, 2^{50}$ **[A1]**

Shows index manipulation to obtain a common power, e.g. $(2^5)^{10}$, $(3^3)^{10}$ and $(5^2)^{10}$ **[M1]**

Evaluates the bases correctly as 32, 27 and 25 **[M1]**

[Back to Question] *[Total Marks: 7]*

11 **(a)** Testing a valid value such as $x = \frac{1}{4}$ to obtain $\frac{1}{16}$, $\frac{1}{2}$, 4, $\frac{1}{4}$ or comparing powers of x **[M1]**

x^2, x, $\sqrt{x}$, $\frac{1}{x}$ **[A1]**

(b) Substituting a valid test value such as $k = -2$ to get -8, $-\frac{1}{2}$, 4, $-\frac{1}{4}$ **[M1]**

Correct ordering of three negative terms: $k^3 < \frac{1}{k} < -\frac{1}{k^2}$ **[M1]**

k^3, $\frac{1}{k}$, $-\frac{1}{k^2}$, k^2 **[A1]**

(c) Rearranging to $n^2 - n \leq 0$ or $n(n-1) \leq 0$ **[M1]**

Identifying critical values $n = 0$ and $n = 1$ **[M1]**

$0 < n \leq 1$ **[A1]**

[Back to Question] *[Total Marks: 8]*

12 0.12 12.1% $\frac{4}{33}$ $\sqrt{0.015}$ **[B2]**

M1 for correct decimal equivalents seen for at least two of the non-decimal values (e.g., 0.121, 0.1212..., 0.1224...).

[Back to Question] *[Total Marks: 2]*

13 Converts all to a common unit, e.g. grams: 0.52 g, 0.04 g, 4.5 g, 0.6 g **[M2]**

(M1 for 2 correct conversions to a common unit)

4.5×10^3 mg, 6×10^{-7} tonnes, 5.2×10^{-4} kg, 0.04 g **[A1]**

[Back to Question] *[Total Marks: 3]*

14 **(a)** B1 for at least two correct evaluated values ($a = 3100$, $b = 3130.49\ldots$, $c = 3125$, $d = 3136$) **[B1]**

B1 for all four evaluated correctly **[B1]**

d, b, c, a **[A1]**

(b) 15 **[B1]**

[Back to Question] *[Total Marks: 4]*

15 **(a)** 5^{-3} 0.00805 $\frac{1}{120}$ 8.5×10^{-3} **[B2]**

If 0 scored, allow [M1] for evaluating at least two correctly as standard decimals: 0.008, 0.0085, 0.00833...

(b) $11.5 \leq m < 12.5$ **[B2]**

If 0 scored, allow [M1] for identifying the bounds 11.5 and 12.5

[Back to Question] *[Total Marks: 4]*

16 **(a)** Converts at least two values to decimals or percentages correctly (e.g. 0.875, 0.86, 0.833...) **[M1]**

$\frac{5}{6}$, 86%, 0.865, $\frac{7}{8}$ **[A1]**

(b) Finds a common denominator (e.g. 6, 12, etc.) **[M1]**

Any correct fraction (e.g. $\frac{5}{12}$, $\frac{2}{5}$) **[A1]**

[Back to Question] *[Total Marks: 4]*

17 **(a)** Evaluating decimals: $p = 0.0048$, $q \approx 0.004878$, $r \approx 0.004796$, $s = 0.00485$, $t = 0.0049$ (at least three correct) **[M1]**

Correct ordering of decimal values **[M1]**

t, q, s, p, r (accept original numerical expressions) **[A1]**

(b) Any correct value in standard form strictly greater than $4.7958\cdots \times 10^{-3}$ and less than or equal to 4.8×10^{-3} (e.g. 4.798×10^{-3} or 4.8×10^{-3}) **[B1]**

(c) Always true **[B1]**

Since $q \approx 0.004878$ and $s = 0.00485$, $q > s$; therefore any $y \geq q$ must also be strictly greater than s **[R1]**

[Back to Question] *[Total Marks: 6]*

18 **(a)** $<$ **[B1]**

(b) Evaluates at least two expressions correctly (0.25, 0.707..., 2) **[M1]**

a^2, a, $\sqrt{a}$, $\frac{1}{a}$ **[A1]**

(c) False **[B1]**

Provides a valid counterexample, e.g. let $p = 2$ and $q = -3$. Then $2 > -3$, but $\frac{1}{2} > -\frac{1}{3}$. **[B1]**

[Back to Question] *[Total Marks: 5]*